

Milestone 4 – Deployment & Final Documentation

AI TraceFinder: Scanner Image Verification System

1. Introduction

Milestone 4 represents the final phase of the TraceFinder project, where the trained deep learning model is deployed through an interactive and visually rich web application. The system enables users to upload scanned images and instantly receive the predicted scanner brand along with a confidence score. To enhance forensic reliability, the application also visualizes scanner-specific artifacts using Grad-CAM heatmaps and maintains a history log of all predictions for traceability.

2. Objectives

- Deploy the trained CNN scanner identification model
- Build a web-based user interface for image upload
- Display predicted scanner brand and confidence score
- Generate explainability using Grad-CAM heatmaps
- Store prediction history with timestamps
- Test the system with multiple sample images
- Document results with screenshots

3. Software & Tools Used

- Python 3.11
- PyTorch
- Torchvision
- Streamlit
- OpenCV
- NumPy
- PIL
- VS Code
- Google Colab

4. System Architecture

1. User uploads a scanned image
2. Image is resized to 224×224 and normalized
3. Preprocessed image is passed to the MobileNetV2 CNN
4. Model predicts scanner brand and confidence score
5. Grad-CAM generates heatmap for explainability
6. Results and logs are displayed and stored in history

5. Model Used

- Architecture: MobileNetV2 (CNN)
- Input size: 224 × 224 × 3

- Output: Multi-class classification
- Classes: Canon120, EpsonV39, EpsonV550, HP, NonScanned
- Model file: `best_scanner_model.pth`

6. Testing Details

The system was tested using multiple scanned and non-scanned images from different sources. Each image was correctly classified with a confidence score, and Grad-CAM successfully highlighted scanner-specific texture and noise regions.

7. Prediction Log

- Predictions are stored in session history
- Fields include:
 - Time
 - Status (Scanned / Not Scanned)
 - Scanner Brand
 - Confidence

8. Limitations

- Performance depends on training data diversity
- Model size is large for lightweight systems
- Currently deployed only on local system

9. Conclusion

Milestone 4 successfully completes the TraceFinder project by deploying a real-time scanner identification system with explainability and logging features. The system demonstrates practical applications in digital forensics, document authentication, and legal evidence verification.